

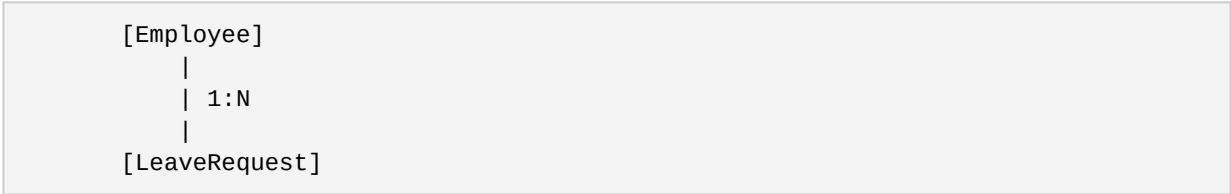
Data Model Documentation

Overview

The system manages the following core domains:

- Employee: Responsible for storing employee details and managing leave requests.
- LeaveRequest: Responsible for tracking leave requests, their statuses, and related metadata.

Entity Relationship Diagram



Entities

Employee

Purpose: Represents an employee within the organization, managing their details and leave balances.

Table/Collection Name: Employee

Fields:

Field Name	Type	Required	Constraints	Description
id	UUID	Yes	Primary key, auto-generated	Unique identifier for the employee
name	VARCHAR	Yes	Max 255 chars	Employee's name
email	VARCHAR	Yes	Max 255 chars, unique	Employee's email
created_at	TIMESTAMP	Yes	Auto-set on create	Record creation time
updated_at	TIMESTAMP	Yes	Auto-set on create/update	Last modification time

Relationships:

- Has many LeaveRequest (1:N): Inverse via LeaveRequest . employee_id

Validation Rules:

- name must be non-empty and unique across all records
- email must be non-empty and unique across all records

LeaveRequest

Purpose: Represents a leave request made by an employee, tracking its status and related metadata.

Table/Collection Name: LeaveRequest

Fields:

Field Name	Type	Required	Constraints	Description
id	UUID	Yes	Primary key, auto-generated	Unique identifier for the leave request
employee_id	UUID	Yes	Foreign key	Employee who requested the leave
start_date	DATE	Yes	-	Start date of the leave
end_date	DATE	Yes	-	End date of the leave
status	VARCHAR	Yes	'PENDING', 'APPROVED', 'REJECTED'	Status of the leave request
created_at	TIMESTAMP	Yes	Auto-set on create	Record creation time
updated_at	TIMESTAMP	Yes	Auto-set on create/update	Last modification time

Relationships:

- Belongs to Employee (N:1): Foreign key employee_id references Employee . id

Validation Rules:

- employee_id must be a valid UUID and reference an existing employee
- start_date must be a valid date and not in the past
- end_date must be a valid date and after start_date

- status must be one of 'PENDING', 'APPROVED', 'REJECTED'

Relationships and Constraints

Foreign Keys

From Entity	Field	To Entity	Field	Cardinality	Delete Behavior
LeaveRequest	employee_id	Employee	id	N:1	CASCADE

Unique Constraints

- Employee(email): Ensures no duplicate email addresses

Data Validation Rules

Business Rules:

- start_date must be before end_date: If start_date is after end_date, return validation error.
- status must be one of 'PENDING', 'APPROVED', 'REJECTED': If status is not one of these values, return validation error.

Format Rules:

- email: Must match the regex `^[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}$`

Data Lifecycle

Creation

- A new Employee is created when a new user is registered.
- Default status for LeaveRequest is PENDING.
- created_by is automatically set to the current user.

Updates

- Employee.email can only be updated by the employee themselves or an admin.
- updated_at is automatically updated on any modification to an Employee or LeaveRequest.

Archival / Soft Delete

- Records are not archived but can be marked as deleted by setting deleted_at timestamp.

- Soft-deleted records are excluded from normal queries.

Indexes

- Primary key: id
- Foreign keys: employee_id
- Search fields: email (for common queries)

Example Records

Employee

```
{
  "id": "550e8400-e29b-41d4-a716-446655440000",
  "name": "John Doe",
  "email": "john.doe@example.com",
  "created_at": "2026-01-15T10:30:00Z",
  "updated_at": "2026-01-15T10:30:00Z"
}
```

LeaveRequest

```
{
  "id": "123e4567-e89b-12d3-a456-426614174000",
  "employee_id": "550e8400-e29b-41d4-a716-446655440000",
  "start_date": "2026-01-15",
  "end_date": "2026-01-20",
  "status": "PENDING",
  "created_at": "2026-01-15T10:30:00Z",
  "updated_at": "2026-01-15T10:30:00Z"
}
```